PREPVECTOR
Cognitive Assessment Platform

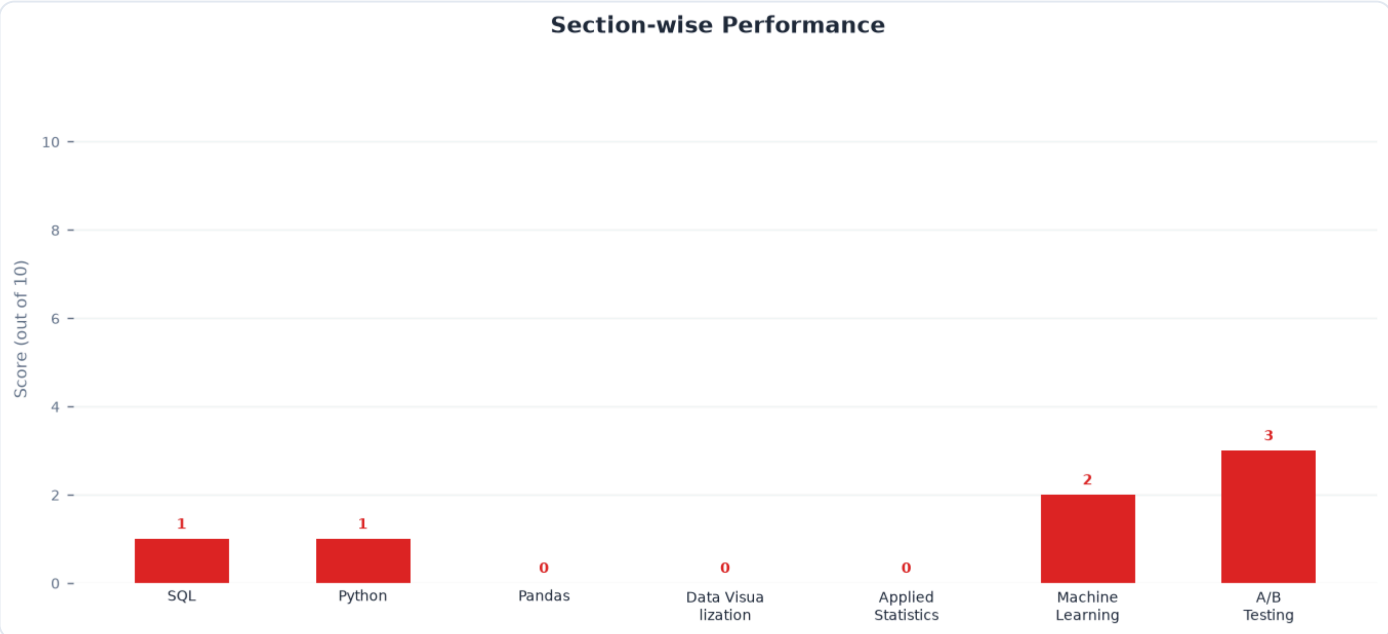
AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE	FINAL SCORE	CORRECT ANSWERS
Alice Helman	10 / 70 weighted points	7 / 35 questions NEEDS IMPROVEMENT

Assessment Date: July 25, 2026 **Report Generated:** July 25, 2026 at 19:32 UTC **Role:** Data Science Expert

OVERALL RATING	SUCCESS RATE	WEIGHTED SCORE	TOP SKILL	NEEDS IMMEDIATE ATTENTION
★☆☆☆☆	14.3%	10 / 70	A/B Testing	Pandas



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	1 / 10	6.8	-5.8	Critical Gap
Python	1 / 10	6.2	-5.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	0 / 10	6.0	-6.0	Critical Gap
Applied Statistics	0 / 10	5.5	-5.5	Critical Gap
Machine Learning	2 / 10	5.2	-3.2	Needs Work
A/B Testing	3 / 10	6.4	-3.4	Needs Work

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	1	4	0	20%
Python	5	5	1	4	0	20%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	0	5	0	0%
Applied Statistics	5	5	0	5	0	0%
Machine Learning	5	5	2	3	0	40%
A/B Testing	5	5	3	2	0	60%

Executive Summary

The candidate achieved a correct-answer count of 7 out of 35 (20.0% accuracy) with a weighted score of 10 out of 70, indicating significant gaps across core data science competencies. While there is a foundational familiarity with Python and SQL, critical technical and conceptual errors prevent successful execution of mid-to-senior level tasks. Immediate, structured upskilling is required to transition from basic syntax to robust analytical execution.

Key Strengths

- **Broad Exposure:** Demonstrates familiarity with advanced concepts like Bayesian updating, SHAP values, and Copy-on-Write in Pandas.
- **Analytical Ambition:** Attempts complex SQL optimizations (correlated subqueries) and advanced Python OOP (descriptors).

Top Blind Spots

- **SQL Aggregations & Window Functions:** Struggles with basic grouping execution, date truncation logic, and the behavioral differences between subqueries and window functions on non-unique keys.
- **Statistical Inference & Experimental Design:** Misinterprets fundamental metrics like Type I/II errors, p-values in hypothesis testing, and critical diagnostic indicators like Sample Ratio Mismatch (SRM) and heteroscedasticity.
- **Python Execution Flow & State:** Exhibits gaps in tracing sequential code execution, particularly with generator state (`yield`), exception handling block order, and Pandas Copy-on-Write memory management.

Actionable Learning Resources

Complete the "SQL for Data Analysis" track on Mode Analytics, focusing specifically on window functions, grouping, and complex aggregations.

✓ Action Item

Read Chapters 2, 3, and 5 of *An Introduction to Statistical Learning (ISLR)* to master linear regression assumptions, hypothesis testing, and bias-variance trade-offs.

✓ Action Item

Work through the "Python Descriptors" and "Generators" deep-dive tutorials on Real Python to master advanced control flow and OOP protocols.

✓ Action Item

Review the official Pandas 2.0+ documentation on Copy-on-Write mechanics and complete practice exercises on DataFrame indexing behavior.

✓ Action Item